

Enhancing Student Performance Prediction Using Machine Learning: A Comparative and Explainable Approach

Research Report

ABSTRACT

Student academic performance prediction can help educational institutions identify performance patterns and support data-driven academic interventions. This study presents a comparative and explainable machine learning framework for predicting students' examination scores using demographic, academic, lifestyle, and behavioral attributes. The proposed framework includes data preprocessing, categorical feature encoding, feature engineering, feature selection, model development, hyperparameter optimization, and model explainability. Multiple machine learning regression models were evaluated using cross-validation and held-out test-set performance metrics, including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), coefficient of determination (R^2), and Mean Absolute Percentage Error (MAPE). After optimization, the XGBoost model achieved an MAE of 3.1881, RMSE of 3.9312, R^2 of 0.9397, and MAPE of 5.13% on the test set. SHAP-based explainability was subsequently applied to interpret the model predictions and identify influential factors. The results indicate that study hours per day and mental health rating were among the most influential features in the prediction model. The study demonstrates the potential of combining predictive machine learning with explainable AI techniques for interpretable student performance analysis.

Keywords: Student Performance Prediction, Machine Learning, XGBoost Regression, Feature Engineering, SHAP, Explainable AI, Educational Data Mining

1. INTRODUCTION

Student academic performance prediction has become an important research area in educational data mining and machine learning. Academic performance is influenced by several factors, including study habits, attendance, lifestyle, mental well-being, and extracurricular activities. Identifying the relationship between these factors and students' examination performance can help educational institutions understand academic patterns and provide appropriate academic support.

Traditional methods of evaluating student performance often rely on historical examination results and descriptive analysis. Although these approaches provide useful information, they may not adequately capture complex relationships among multiple academic, demographic, and behavioral factors. Machine learning techniques can address this limitation by learning patterns from historical student data and using them to predict examination performance.

Several regression algorithms can be applied to student performance prediction, but their predictive capabilities may differ depending on the characteristics of the dataset. Therefore, comparative evaluation of multiple machine learning models is important. In addition to predictive accuracy, model interpretability is important in educational applications because educators need to understand which factors contribute to predictions.

This study presents a comparative and explainable machine learning framework for predicting students' examination scores. The framework includes data preprocessing, categorical feature encoding, feature engineering, feature selection, model development, hyperparameter optimization, and model evaluation. Multiple regression models are evaluated using five-fold cross-validation and an independent test set using MAE, RMSE, R^2 , and MAPE.

The study further applies SHAP (SHapley Additive exPlanations) to interpret the predictions of the optimized XGBoost model. This enables identification of influential features and provides insights into how different student-related factors contribute to predicted examination scores.

2. RELATED WORK

Student performance prediction has been widely studied using machine learning and educational data mining techniques. Researchers have explored academic, demographic, behavioral, and lifestyle-related factors to identify patterns associated with student achievement.

Traditional statistical approaches such as linear regression have been used to model examination performance. However, student performance can involve nonlinear relationships and interactions among multiple factors. Consequently, decision trees, random forests, support vector regression, and gradient boosting methods have been investigated.

Feature engineering and feature selection can improve predictive performance by reducing unnecessary information and improving data representation. Ensemble-based methods, particularly gradient boosting algorithms, are effective approaches for structured tabular datasets.

Interpretability is increasingly important when machine learning is applied to educational data. Explainable Artificial Intelligence techniques can help researchers and educators understand factors contributing to predictions. SHAP is one approach that provides global and local explanations.

The present study combines comparative machine learning evaluation with explainability. Multiple regression models are evaluated using cross-validation and independent test-set metrics, followed by hyperparameter optimization and SHAP analysis.

3. METHODOLOGY

3.1 Dataset Description

The dataset contains 1,000 student records and 16 original features related to demographic, academic, lifestyle, and behavioral characteristics. The target variable is the students' examination score. The dataset was divided into 80% training and 20% testing, giving 800 training observations and 200 test observations.

3.2 Data Preprocessing

Data preprocessing was performed before model development. Numerical and categorical variables were handled separately, with categorical variables encoded using one-hot encoding. Feature engineering was subsequently performed. Two engineered features, `academic_engagement` and `study_attendance_interaction`, were identified as highly redundant with their original components and removed. After feature reduction, the dataset contained 14 features for the reduced-feature modelling stage.

3.3 Feature Engineering and Selection

Feature engineering was used to improve the representation of student-related information. Feature selection was performed to reduce redundancy and improve model efficiency. Lasso regression was also evaluated as a regularized feature-selection approach. The optimized Lasso model was obtained with a best alpha value of 0.1.

3.4 Machine Learning Models

Multiple regression models were evaluated for predicting examination scores. The optimized XGBoost regression model used the following parameters: `n_estimators=500`, `max_depth=2`, `learning_rate=0.05`, `min_child_weight=1`, `subsample=0.9`, `colsample_bytree=0.8`, and `random_state=42`.

3.5 Model Evaluation

Model performance was evaluated using MAE, RMSE, R^2 , and MAPE. Five-fold cross-validation was used during model comparison and hyperparameter optimization, while the final model was evaluated on the independent test set.

3.6 Model Explainability

SHAP (SHapley Additive exPlanations) was applied to the optimized XGBoost model. Mean absolute SHAP values were used for global feature importance, while SHAP summary and dependence plots were used to investigate feature contributions and relationships.

4. RESULTS AND DISCUSSION

4.1 Five-Fold Cross-Validation Results

For optimized Lasso, the best alpha was 0.1 with a cross-validation RMSE of 5.3928. For optimized XGBoost, the best parameter configuration resulted in a cross-validation RMSE of 5.5146.

Table 1. Five-Fold Cross-Validation Results

Model	Best Hyperparameter	CV RMSE
Lasso	$\alpha = 0.1$	5.3928
XGBoost	<code>n_estimators=500</code> ; <code>max_depth=2</code> ; <code>learning_rate=0.05</code>	5.5146

4.2 Independent Test-Set Performance

The optimized Lasso model achieved MAE 4.0620, RMSE 5.0126, R^2 0.9020, and MAPE 6.73%. The optimized XGBoost model achieved MAE 3.1881, RMSE 3.9312, R^2 0.9397, and MAPE 5.13%.

Table 2. Independent Test-Set Performance

Model	MAE	RMSE	R ²	MAPE
Optimized Lasso	4.0620	5.0126	0.9020	6.73%
Optimized XGBoost	3.1881	3.9312	0.9397	5.13%

The optimized XGBoost model produced lower MAE, RMSE, and MAPE and higher R² than optimized Lasso on the independent test set. Therefore, XGBoost was selected as the final predictive model for subsequent error analysis and SHAP-based interpretation.

4.3 Actual versus Predicted Performance

The actual-versus-predicted plot for optimized XGBoost shows strong alignment between predicted and actual examination scores. Most observations are close to the reference relationship, while a smaller number show larger deviations.

Figure 1. Actual versus Predicted Examination Scores. [Insert the original Colab plot here.]

4.4 Residual Analysis

The residual histogram is centered close to zero. Most residuals occur within a moderate range, although a small number of observations show comparatively larger errors.

Figure 2. Distribution of XGBoost Prediction Residuals. [Insert the original Colab plot here.]

4.5 Error Analysis

The largest individual prediction errors were inspected using the absolute error between actual and predicted examination scores.

Table 3. Ten Largest Prediction Errors

Index	Actual	Predicted	Residual	Absolute Error
121	71.1	83.0596	-11.9596	11.9596
91	58.6	49.0687	9.5313	9.5313
45	71.4	80.8014	-9.4014	9.4014
99	87.8	78.6238	9.1762	9.1762
65	37.4	46.2002	-8.8002	8.8002
19	59.4	51.3669	8.0331	8.0331
92	44.1	52.1186	-8.0186	8.0186
10	53.0	45.0608	7.9392	7.9392
133	76.1	68.5350	7.5650	7.5650
29	59.9	67.2371	-7.3371	7.3371

The largest absolute error was observed for index 121, where the actual score was 71.1 and the predicted score was 83.0596, resulting in an absolute error of 11.9596. The listed high-error observations include both overprediction and underprediction.

4.6 Feature Importance

The optimized XGBoost feature-importance analysis identified study hours per day as the most important transformed feature, followed by mental health rating and exercise frequency. Social media hours, Netflix hours, sleep hours, and attendance percentage also contributed.

Table 4. Top 15 XGBoost Feature Importance Values

Rank	Feature	Importance
1	num__study_hours_per_day	0.508771
2	num__mental_health_rating	0.161717
3	num__exercise_frequency	0.063241

4	num_social_media_hours	0.050916
5	num_netflix_hours	0.048177
6	num_sleep_hours	0.035522
7	num_attendance_percentage	0.016810
8	cat_gender_Male	0.010792
9	cat_extracurricular_participation_Yes	0.009588
10	cat_internet_quality_Good	0.009348
11	cat_diet_quality_Good	0.008236
12	cat_gender_Female	0.007864
13	cat_parental_education_level_Master	0.007562
14	cat_parental_education_level_High School	0.007515
15	cat_internet_quality_Average	0.007273

Figure 3. Top 15 Features — Optimized XGBoost. [Insert the original Colab bar chart here.]

4.7 SHAP Explainability

SHAP analysis was used to explain the optimized XGBoost predictions. Study hours per day had the largest mean absolute SHAP value, followed by mental health rating, social media hours, exercise frequency, Netflix hours, sleep hours, and attendance percentage.

Table 5. Top 15 Features Based on Mean Absolute SHAP Value

Rank	Feature	Mean Absolute SHAP Value
1	num_study_hours_per_day	11.322068
2	num_mental_health_rating	4.736582
3	num_social_media_hours	2.466869
4	num_exercise_frequency	2.204530
5	num_netflix_hours	2.061608
6	num_sleep_hours	2.022720
7	num_attendance_percentage	1.101787
8	cat_diet_quality_Fair	0.198754
9	cat_diet_quality_Good	0.164238
10	num_age	0.154629
11	cat_internet_quality_Average	0.114806
12	cat_internet_quality_Good	0.045505
13	cat_internet_quality_Poor	0.039178
14	cat_parental_education_level_Bachelor	0.037048
15	cat_parental_education_level_High School	0.036576

Figure 4. SHAP Summary Plot — Optimized XGBoost. [Insert the original Colab plot here.]

Figure 5. Top 15 Features Based on Mean Absolute SHAP Value. [Insert the original Colab plot here.]

4.8 SHAP Dependence Analysis

The study-hours dependence plot shows a strong increasing relationship between study hours and SHAP contribution, indicating that higher study-hour values generally contribute positively to the predicted score while lower values contribute negatively.

The mental-health-rating dependence plot shows an increasing pattern, with higher standardized mental health rating values associated with increasingly positive SHAP contributions. The exercise-frequency plot shows a similar positive relationship.

The sleep-hours dependence plot shows an overall positive relationship between sleep hours and SHAP contribution. In contrast, the social-media-hours dependence plot shows an overall negative relationship over a substantial portion of its observed range.

Figure 6. SHAP Dependence Plot — Study Hours per Day. [Insert the original Colab plot here.]

Figure 7. SHAP Dependence Plot — Mental Health Rating. [Insert the original Colab plot here.]

Figure 8. SHAP Dependence Plot — Exercise Frequency. [Insert the original Colab plot here.]

Figure 9. SHAP Dependence Plot — Sleep Hours. [Insert the original Colab plot here.]

Figure 10. SHAP Dependence Plot — Social Media Hours. [Insert the original Colab plot here.]

5. FINAL MODEL SELECTION

Although optimized Lasso achieved a slightly lower five-fold cross-validation RMSE of 5.3928 than optimized XGBoost at 5.5146, the independent test-set results showed stronger predictive performance for XGBoost. XGBoost achieved MAE 3.1881, RMSE 3.9312, R^2 0.9397, and MAPE 5.13%, compared with 4.0620, 5.0126, 0.9020, and 6.73% for Lasso. SHAP analysis also provided an interpretable explanation of the XGBoost predictions. Therefore, optimized XGBoost was used as the final model.

6. DISCUSSION

The experimental results demonstrate that machine learning can model student examination scores from academic, lifestyle, behavioral, and demographic variables. The optimized XGBoost model achieved an R^2 of 0.9397 on the independent test set.

Feature-importance and SHAP analyses consistently identified study hours per day as the dominant feature. Mental health rating was the second most influential feature. Exercise frequency, social media hours, Netflix hours, sleep hours, and attendance percentage also contributed.

The SHAP dependence plots provided information beyond feature ranking. Study hours, mental health rating, exercise frequency, and sleep hours showed generally increasing SHAP contributions, whereas social media hours showed an overall decreasing pattern over a substantial part of the observed range.

The error analysis showed that most predictions were reasonably close to actual values, while a smaller number had comparatively larger errors.

7. LIMITATIONS

The study is based on 1,000 student records, and observed relationships may depend on the characteristics of this dataset.

The analysis is predictive rather than causal. Feature importance and SHAP contributions describe how the trained model uses variables for prediction and do not establish causal effects.

External validation on an independent dataset from another institution would provide additional evidence of generalizability.

Categorical variables are represented through encoded features, so individual encoded categories should be interpreted in the context of the complete categorical variable.

8. CONCLUSION

This study developed a comparative and explainable machine learning framework for predicting student examination performance. The workflow included data preprocessing, feature engineering, feature reduction, model evaluation, hyperparameter optimization, and SHAP-based explainability. Optimized Lasso and XGBoost were evaluated using five-fold cross-validation and an independent test set.

The optimized XGBoost model achieved an MAE of 3.1881, RMSE of 3.9312, R^2 of 0.9397, and MAPE of 5.13% on the test set. SHAP analysis showed that study hours per day and mental health rating were the most influential features among the evaluated transformed variables. Combining predictive performance with model explainability provides an informative framework for student performance analysis.

The findings can support further research using larger datasets, external validation, and additional explainability analyses. The results should be interpreted as model-based associations rather than causal conclusions.

9. REFERENCES

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